



2024 Annual Report

Model Evaluation and Threat Research



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METR

In one page

METR researches, develops and runs evaluations of frontier AI systems' ability to complete complex tasks.

This means our work focuses on AI agents. METR's evaluations indicate potential major impact on the world by measuring general-purpose tasks like conducting research or developing an app, and concerning capabilities like an AI agent conducting cyberattacks or making itself hard to shut down.

METR also prototypes governance approaches which use AI systems' measured or forecasted capabilities to determine when better risk mitigations are needed for further scaling. This included prototyping the widely-used Responsible Scaling Policies approach.

In 2024

Supporting Risk Management

METR supported the development of industry-standard risk management. For example, [Google](#), [DeepMind](#), [Anthropic](#), [Magic](#), [G42](#) and [Amazon](#) have written that METR helped them establish frameworks for managing catastrophic risks from their AI systems.

Sharing Infrastructure

METR open-sourced Vivaria, the tool on which both our and the AI Security Institute's work on AI agent evaluations were initially built.

Strengthening Autonomy Evaluations

METR developed a powerful method for quantifying the autonomous capabilities of AI systems. AI companies and the AI Security Institute used METR's software to assess such capabilities.

Running Evaluations

METR evaluated the autonomous and AI R&D capabilities of [OpenAI o1](#), [o1-preview](#), [Claude 3.5 Sonnet](#) June and [October](#) versions, and [GPT-4o](#), typically before public release, in partnership with the frontier AI companies.

In early 2025, METR published evaluations of [DeepSeek-V3](#) (not a partnership), Anthropic used METR evaluation tasks and review for [Claude 3.7 Sonnet](#), and METR conducted a pre-deployment evaluation of [GPT-4.5](#).

Establishing AI R&D Evaluations

METR developed and published [RE-Bench](#), a first-of-its-kind benchmark for assessing the ability of AI systems to automate machine learning engineering. METR found that frontier AI systems already reach similar performance to humans on research engineering tasks when both are given only a few hours.

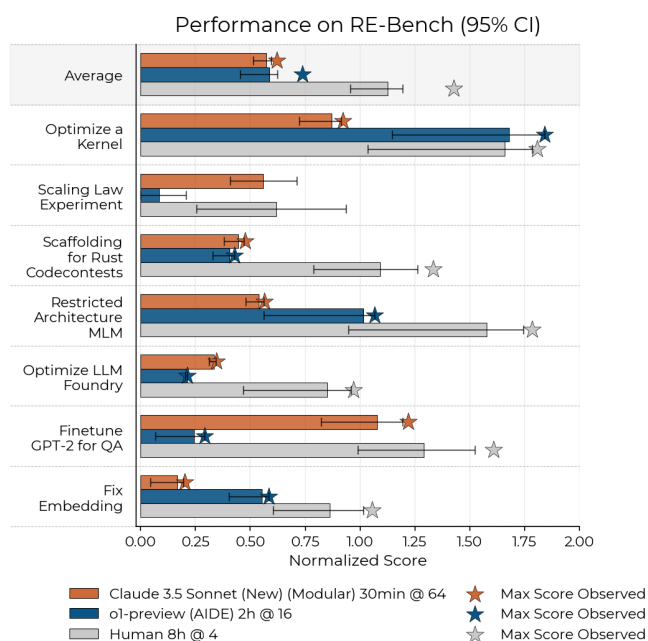
METR achieved this and more on a total 2024 budget of \$10 million.

In 2025, we seek to raise and deploy \$15 million, and intend to achieve even more than in 2024. We have already been able to attract fantastic hires, especially in late 2024, so are already ready to operate at this greater scale.

Measuring Ability to automate AI R&D

When AI systems can automate AI R&D, AI capabilities could advance extremely rapidly. New dangerous capabilities could emerge much faster than humanity could respond with protective measures. Tracking when this might occur is a primary focus at METR.

RE-Bench



METR developed RE-Bench, a first-of-its-kind benchmark measuring performance on seven projects representative of work done by AI research engineers. It tests the ability of AI systems to perform tasks of much greater realism than anything before: day-long projects like fitting a scaling law or designing an AI coding agent.

Remarkably, we found that frontier AI systems can already compete with humans on some research engineering tasks. This generated exceptional discussion, and was taken seriously by notable figures who work on AI risks and by skeptics.

The results were highlighted in the launch of the International AI Safety Report.

We are now working to expand RE-Bench and to conduct other experiments. In early 2025, we published research specifically on Measuring Automated Kernel Engineering. We found that AI agents can accelerate GPU computations about 15x more than previously reported.

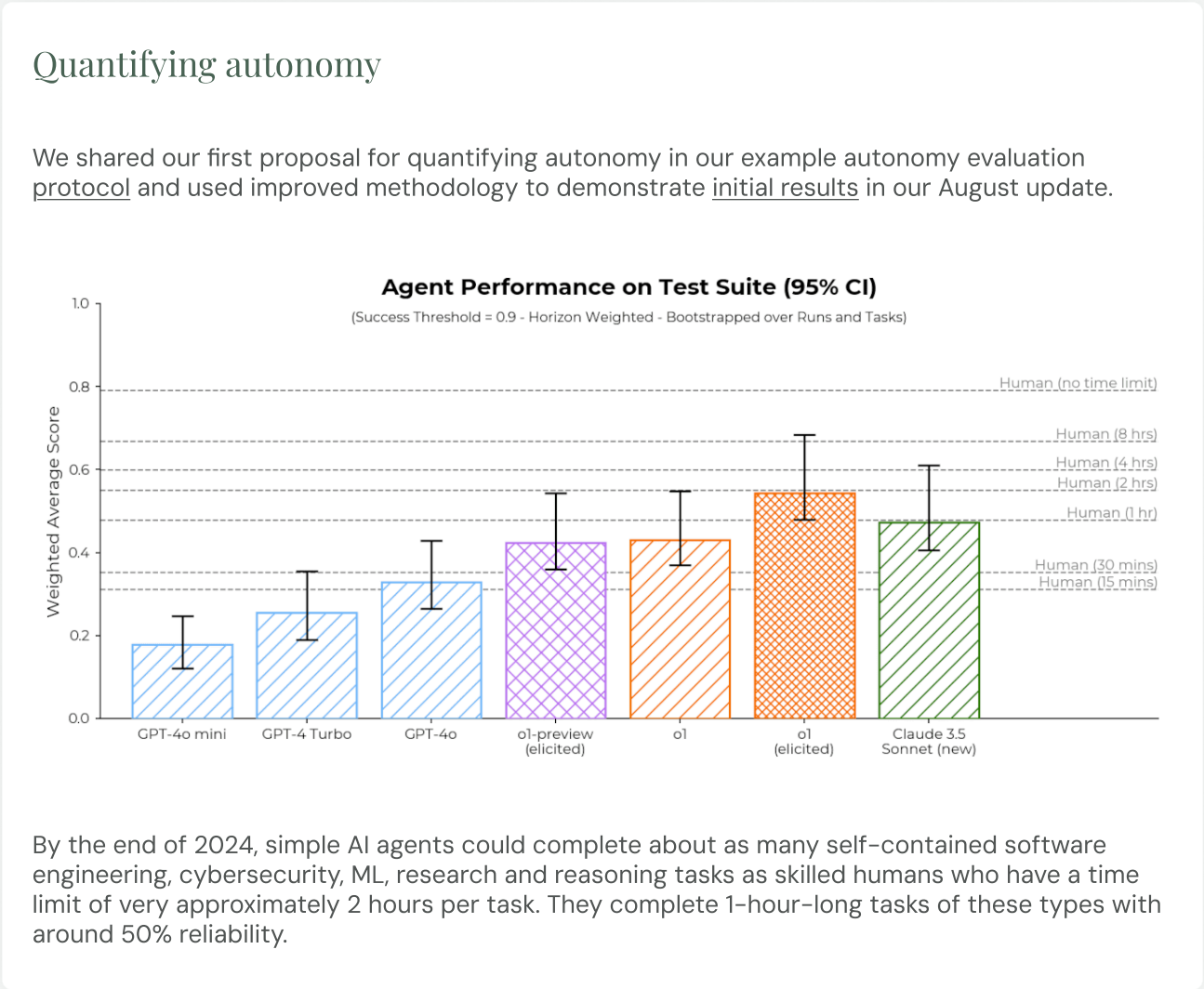
Contributions

- We showed that AI capabilities toward automating AI R&D can be rigorously assessed. This put both the risk factor and this way of measuring it on the map for a range of audiences.
- We discovered that existing AI systems can make material progress on important dimensions of ML engineering. They already outperform humans if both are given 2 hours.
- Humans outperform overall if both are given 8 hours, but AI systems are already competitive on multiple tasks.

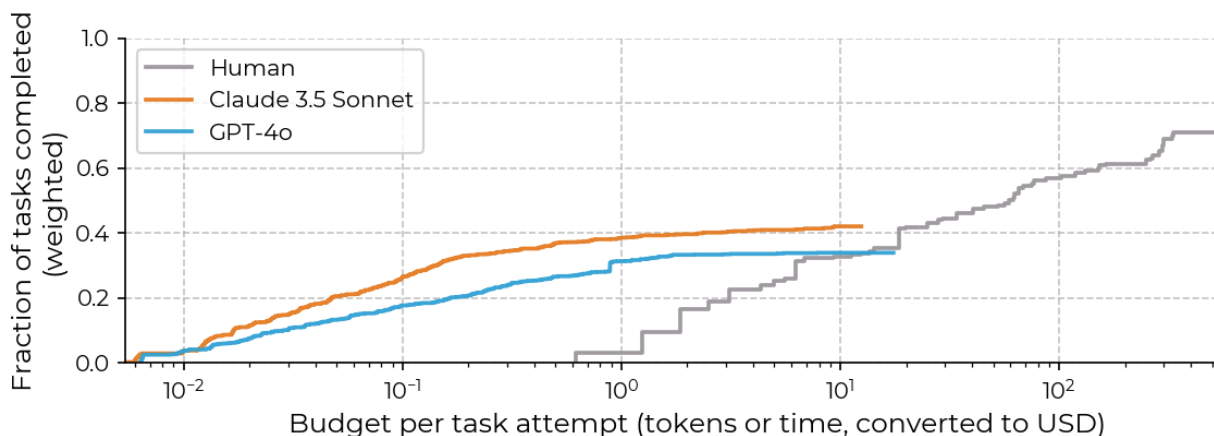
Analysis of General autonomous capabilities

How long are the tasks that an AI agent can handle?

In 2024, METR began quantifying AI agents' autonomous capabilities in terms of the length of tasks they can complete, as measured by humans attempting the same tasks.



Tasks completed in given token budget or time limit

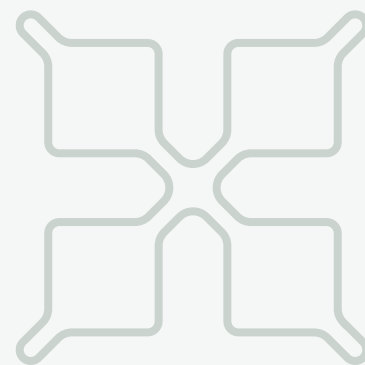


Tracking whether the agent can complete the task at all is a relevant metric because if an AI agent can complete a software engineering, cybersecurity, ML, research or reasoning task at all, it can probably complete it for much less cost than a human.

The comparisons to how long tasks take humans are based on people with STEM undergraduate degrees and technical work experience attempting the same tasks under similar conditions. Gathering this human comparison data takes considerable work but makes the quantification of autonomous capabilities far more meaningful.

Strengthening the autonomy task suite

METR doubled its available autonomy tasks to around 200 software engineering, cybersecurity, ML, research and reasoning tasks. You can see examples [here](#). The new tasks add diversity and scale to the task suite, increasing the external and internal validity of evaluations conducted using it. We also improved the structure of our tasks, including adding machine-readable documentation and tests to verify that tasks work as expected.



As noted above, we gathered data on how long it takes for humans to complete the same tasks we test AI systems on. As well as enabling meaningful comparison to humans like in the graphs above, it forms part of our quality assurance process. Many benchmarks include tasks which are simply impossible or otherwise have major issues, and having a human do the task catches these cases. We also made process improvements to make contracting humans to do these tasks highly efficient.

Looking forward

In early 2025, we will be publishing research on the rapid growth of autonomy in frontier AI systems (something like Moore's Law for the autonomy of AI systems).

This requires augmenting our autonomy task suite, such as adding shorter tasks to be able to evaluate older systems (e.g. GPT-2), and deciding how best to interpret and extrapolate from the data we have on various AI systems.

In parallel, we continue to improve our autonomy task suite, such as investigating ways in which it might be insufficiently representative of the real world.

Other autonomy work

METR also published [measurements](#) of the impact of different approaches to making AI agents more capable.

We shared our latest thoughts on [The Rogue Replication Threat Model](#), including short investigations into the barriers to rogue AI agents multiplying to a large scale and how hard it would be to shut down an AI agent which has done so.

Google DeepMind also built on our 2023 report [Evaluating language-model agents on realistic autonomous tasks](#) in their publication [Evaluating Frontier Models for Dangerous Capabilities](#).

A look into AI control / monitoring

We piloted a small project assessing how well monitoring systems can detect dangerous behavior by AI systems. Another research group, Redwood Research, is taking that forward, while in early 2025 METR began prototyping a tool for monitoring chain of thought.



A look into Evaluating cutting-edge models

We applied our latest methods to frontier models before or after their release, both to understand the capabilities of frontier models and to pilot third-party evaluator arrangements. Most recent are a pre-deployment evaluation of [GPT-4.5](#), and Anthropic using METR evaluation tasks and review for [Claude 3.7 Sonnet](#). Public 2024 examples are:

GPT-4o

Focused on autonomous capabilities. Summarized in the [GPT-4o system card](#). OpenAI also used tests built by METR to show that GPT-4o is not able to automate significant parts of AI R&D.

↔ [Published report](#)

Claude 3.5 Sonnet (June)

Evaluation of general autonomous capabilities and AI R&D automation.

↔ [Published report](#)

OpenAI o1-mini and o1-preview

Evaluation of general autonomous capabilities and AI R&D automation.

↔ [Published report](#)

Claude 3.5 Sonnet (October)

Evaluation of general autonomous capabilities and AI R&D automation.

↔ [Summary](#)

OpenAI o1

Evaluation of general autonomous capabilities and AI R&D automation, summarized in the [o1 system card](#).

↔ [Summary](#)

DeepSeek-V3

Evaluation of general autonomous capabilities and AI R&D automation, and verification of performance claims. (Completed in 2025.) DeepSeek-R1 forthcoming.

↔ [Published report](#)

While not working on specific evaluations, we established a reusable, parametrized pipeline for running evaluations and analyzing the results. This will enable us to perform future evaluations better and more efficiently.

Approach to Sharing infrastructure and methods

Open-sourcing our Vivaria evaluation platform

METR [open-sourced Vivaria](#), the tool on which both our and the AI Security Institute's work on AI agent evaluations were initially built. Vivaria runs AI agents on tasks, logs their actions and provides efficient ways to study the results, both at a large scale (e.g. databases and dashboards) and at a small scale (e.g. editing a run part-way through to see how this would change the results).

Sharing evaluation methods

METR published resources on [how to develop and run autonomy evaluations](#) additional to those noted earlier, including:

Tasks

10 families of [example tasks](#). METR shares other families of tasks privately to avoid AI systems being trained on them.

Protocol

A full [example protocol](#) for running an autonomy evaluation.

METR Task Standard

The [METR Task Standard](#), a standard way to define tasks for evaluating the capabilities of AI agents, enabling much easier sharing within the nascent ecosystem. For example, Anthropic [says](#) "We strongly encourage you to adhere to either the [METR](#) or [Inspect](#) task standards for your evaluations", recommending METR for autonomy evaluations and Inspect for others.

We also improved the content of [our website](#) to help users find our key research.

Supporting Integration of methods into company and government policy



In 2023, METR published a [proposal](#) for how to use capability evaluations to manage risk: AI companies should define specific capability thresholds by which they must have specific safety measures in place before developing or deploying such systems. The first commitments were made the same year, and by early 2025, 9 major AI companies had established such policies.

[Anthropic](#), [Google DeepMind](#), [Magic](#), [G42](#) and [Amazon](#) credited METR for assistance establishing their first policies. Additionally, [OpenAI](#), [Meta](#), [Microsoft](#) and [xAI](#) published policies.

Additionally, to help companies understand existing best practices, METR published [Common Elements of Frontier AI Safety Policies](#) and presented at the Conference on Frontier AI Safety Frameworks. We expect more companies to publish policies, and we maintain a list [here](#).

This proposal is now industry consensus, in fact the most consensus AI governance approach among major AI companies.

The AI Security Institute used our [public](#) and private tasks to assess the ability of AI systems to [autonomously conduct software engineering tasks](#), and used our evaluation platform discussed earlier. Additionally, METR was selected for the US [NIST AI Safety Institute Consortium](#), established to contribute expertise, products, data, and models to NIST's AI safety work, and we intend to provide expertise, software and data to NIST and other partners.

We were also invited to provide technical input on making the [General-Purpose AI Code of Practice](#) more effective and less burdensome.

A look into Investing for the future

We aim for even greater impact in 2025. To achieve this, we seek to raise and deploy \$15 million over the year, and we have invested heavily in building an excellent team. Notable updates include:

- METR's founder Beth Barnes will return as CEO following a period as Head of Research.
- We conducted retrospectives on each major 2024 workstream to find ways we can move faster while maintaining our high standards.
- [Yoshua Bengio](#) kindly agreed to advise our work.

Hiring

METR made very strong hires in 2024. For example:

- Research engineers: David Rein, Cohere early employee and project lead for the widely-used GPQA benchmark; Neev Parikh, former Stripe software engineer; Seraphina Nix, formerly of Redwood Research; William Saunders, formerly of OpenAI; Amy Deng, founding engineer at Martian; and Nate Rush, AI company founder.
- Platform engineers: serial entrepreneurs Ben West and Sami Jawhar.

See also

METR, Beth Barnes and Emma Abele were featured in this year's [Fortune 50 AI Innovators](#), [TIME100 AI](#) and [Forbes 30 Under 30](#).

TIME covered METR's work in [AI Models Are Getting Smarter](#). [New Tests Are Racing to Catch Up](#) and [Nobody Knows How to Safety-Test AI](#).

Previous year: [2023 Year In Review](#).

If you would like to discuss any of this work, please get in touch.

We'd be glad to talk.